Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	System currently runs on a local Flask server	Limited accessibility for multiple users	High
2	Uses a static CSV dataset	Dataset updates require manual intervention	Medium
3	No real-time weather integration	Predictions rely only on user-entered values	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users on localhost	Deploy on cloud platform to support thousands of users
2	Data Storage	CSV-based dataset storage	Use MySQL/PostgreSQL database for efficient storage and retrieval
3	Performance	Local model prediction	Optimize model and use cloud infrastructure for faster predictions
4	Security	Basic application security	Implement authentication, HTTPS, and secure data storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy application on cloud platform	3 Months	Improved accessibility and scalability
Phase 2	Real-time weather API integration	6 Months	More accurate crop recommendations
Phase 3	Fertilizer recommendation system	9 Months	Better crop productivity and soil management
Phase 4	Mobile application for farmers	12 Months	Increased usability and wider adoption